BUSINESS PROCESS MANAGEMENT, COURSE PROJECT

Detecting and reducing unnecessary complexity in business process models

Understanding of the topic, a clear problem statement, and three base papers, explained from the start in plain language.

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READ THIS FIRST

The whole project in one page

Your course asks for three things for now. This document gives you all three, and also a short guide to presenting them. Each part starts from zero, so you can read it in order even if you are new to the subject.

YOU NEED TO PRESENT	WHERE TO FIND IT	IN ONE LINE
1. Understanding of the topic	Part 1	A process model is a diagram of how work is done. Some models are more tangled than the process needs. That extra tangle is unnecessary complexity.
2. Clear problem statement	Part 2	We want to find unnecessary complexity in a model using simple measurements and checks, and remove it without changing what the process does.
3. Base papers and references	Part 3	Three papers: two literature reviews (2018 and 2022) and one classic set of seven modelling guidelines (2010). All are cited by other researchers and all are readable at 3rd year level.

Short version for a nervous presenter. A business process model is useful only if people can read it. Research shows that models with many elements, many branching paths and a tangled structure are harder to understand and more likely to contain mistakes. Researchers have proposed measurements (metrics) to detect such models and rules (guidelines) to fix them. Our project collects these ideas, applies them to a small set of real models, and checks whether the fixes really make the models simpler.

How to read this document

1. **Part 1** teaches the topic step by step: process, process model, complexity, how to detect it, how to reduce it. It uses one running example that you can reuse in your slides.
2. **Part 2** turns the topic into a problem statement, objectives and research questions. You can edit these with your teammate and your professor.
3. **Part 3** explains why each paper was chosen, what it says in simple words, and how you will use it.
4. **Part 4** gives a slide plan, a way to split the talk between two people, and answers to questions a professor is likely to ask.

A note on honesty. Everything stated as a finding from a paper comes from the abstract or full text that was read while preparing this document. Where a paper was only checked by title, venue and citation count, this is said clearly in Part 3.

PART 1

Understanding the topic

1.1 Start with the basics: what is a business process

A **business process** is a series of steps that people, software, or both carry out to reach a goal that matters to a customer or to the organisation. You meet business processes every day:

- A customer orders a pizza online and it arrives at the door.
- A student asks the college office for a bonafide certificate.
- A bank checks a loan application and approves or rejects it.

Every process has the same basic parts. Something **triggers** it (an order arrives). There is **work** to do (check ingredients, cook, deliver). There are **decisions** (is the item available or not). There are **people or systems** doing each step. And there is an **outcome** (the customer gets the pizza).

1.2 What is business process management

Business process management (BPM) is the practice of understanding, improving and running processes on purpose instead of by habit. A company that does BPM usually moves through a cycle:

1. **Identify** which processes matter most.
2. **Discover** how each one works today and draw it. This drawing is the "as-is" model.
3. **Analyse** the model to find problems such as delays, rework and mistakes.
4. **Redesign** the process to fix those problems. This gives a "to-be" model.
5. **Implement** the change, often with software that runs the model.
6. **Monitor** the results and start the cycle again.

Notice that the **model** sits in the middle of steps 2 to 5. People draw it, discuss it, analyse it and sometimes feed it into software. If the model is hard to read or wrong, every later step suffers. That is why the quality of a model matters, and it is the starting point of our topic.

If your course follows the textbook Fundamentals of Business Process Management by Dumas, La Rosa, Mendling and Reijers, this is the same cycle that book uses.

1.3 What is a business process model

A **process model** is a diagram that shows the steps of a process and the order in which they happen. The most widely used drawing language is **BPMN**, short for Business Process Model and Notation. It is an international standard, so people in different companies can read each other's diagrams.

You only need a handful of BPMN symbols for this project. Figure 1 shows them.



Figure 1. The BPMN symbols used in this document. Arrows show the order of work.

What the symbols mean

SYMBOL	MEANING
Start event	The thin circle. The process begins here, for example "order received".
End event	The thick circle. A path of the process finishes here.
Task	The rounded box. One piece of work, named with a verb and an object, such as "Pack order".
Gateway	The diamond. A point where the path splits or comes back together. A **split** sends the flow into several paths. A **join** brings paths back into one.
XOR gateway	Diamond with an X. Exactly one path is taken ("either this or that"). Example: item in stock or not in stock.
AND gateway	Diamond with a plus. All paths are taken at the same time. Example: pack the order and print the invoice together.
OR gateway	Diamond with a circle. One or more paths are taken, in any combination. It is the hardest one to reason about.

1.4 What does "complexity" mean for a process model

In everyday words, the **complexity** of a model is how hard it is to read, to check for mistakes, and to change. It comes from two different places, and this difference is the heart of the topic.

- **Essential complexity** comes from the real process. A loan approval with twelve different rules really is complicated, and the model has to show that.

- **Unnecessary complexity** comes from how the model was drawn. It could be removed without losing any information and without changing what the process does.

Think of a city map. A large city needs a large map, and that is essential complexity. A map that draws the same street twice, or uses a different symbol for every bus stop, is harder to use for no good reason. That is unnecessary complexity.

> **Working definition for our project.** Unnecessary complexity is the part of a process model's size and tangle that is caused by modelling choices rather than by the real process, and that can be reduced while the model keeps showing the same behaviour.
>
> This wording is ours, built for the project. The idea of splitting essential from avoidable complexity is common in software engineering. The papers in Part 3 use related terms such as structural complexity.

Examples: essential or unnecessary?

SITUATION IN A MODEL	TYPE	WHY
A loan process with twelve genuine approval rules	Essential	The real process is rich. The model must show it.
Two tasks that really happen in parallel, drawn with an AND gateway	Essential	The parallel work is part of the process.
The same task "Register order" drawn twice on two separate paths	Unnecessary	One copy, reached from both starts, says the same thing.
A diagram with 120 elements on a single page, where groups of tasks could sit inside collapsed sub-processes	Unnecessary at that level	A reader does not need all the detail at once.
A splitting gateway with no matching joining gateway	Unnecessary	It makes the flow hard to follow and easy to get wrong.

In practice the line between the two is blurry, and people disagree about where it falls. This is one reason the topic is a research problem and not just a drawing tip.

1.5 Where unnecessary complexity comes from

Researchers and practitioners keep finding the same habits that inflate models. The table lists the common ones. The last column links each habit to a guideline from Paper 3 in Part 3, so you can see that the advice in the literature lines up with the symptoms.

Common causes of unnecessary complexity

HABIT	WHAT IT LOOKS LIKE	GUIDELINE
Too many elements	Tasks, events and gateways that add nothing, or detail that is not needed at this level.	G1, G7
Gateways with many branches	One diamond that sends the flow into five or six paths.	G2
Several start or end events	Two or three starts and four ends, so a reader must check every combination.	G3
Unstructured flow	Splits that never meet a matching join, paths that jump around. People call this a spaghetti model.	G4
OR gateways	Paths where "one or more" can be taken. The meaning is easy to misread.	G5
Unclear labels	Task names such as "Order handling" that do not say what is done.	G6
Everything on one level	A very large diagram with no sub-processes.	G7
Repeated fragments	The same group of steps copied into several places.	Refactoring work (Part 3, extra reading)

1.6 Why it matters

It is fair to ask whether this is only about neat drawings. The research says it is more than that.

- **People understand the model less well.** A systematic review of empirical studies (Paper 1) found that structural complexity is the factor on which researchers agree most: it makes models harder to understand.
- **Models contain more mistakes.** The authors of Paper 3 link larger and more tangled models to higher error rates, such as paths where the process could get stuck. They report that for very large models, with more than 50 elements, the chance of an error tends to go above 50%.
- **Mined models become useless.** When models are discovered automatically from system logs (process mining), real processes are often unstructured, so the models come out complex and give little practical insight. This is the starting point of Paper 2.
- **Cost follows.** A confusing model takes longer to explain to new people, is risky to change, and is a poor base for automating a process. This last point is our own reasoning, not a quoted finding.

The people who draw models are often not trained modellers. Paper 3 points out that many casual modellers have limited modelling skill, which is exactly why simple, evidence-based rules are useful.

1.7 Detecting unnecessary complexity

To reduce something, we first need to see it. There are three ways to detect it, and a research project can use more than one.

1. **Measure the model with metrics.** A metric is a number computed from the diagram, such as how many elements it has.
2. **Check the model against rules.** A rule says "a model should have one start event". A script can test it and report each place where the rule is broken.
3. **Ask people.** Give readers a model and a few questions, and see how fast and how correctly they answer. Many of the studies behind Papers 1 and 3 work like this.

Common metrics

METRIC	WHAT IT COUNTS	IDEA BEHIND IT
Size	Number of nodes (events, tasks, gateways)	More elements means more to read and more places for mistakes.
Number of arcs	Number of arrows between nodes	More arrows means more paths to follow.
Gateway count and mix	How many gateways, and of which type	Gateways are where reading gets hard. OR gateways are the hardest.
Largest fan-out	The most branches leaving a single gateway	A diamond with six exits is much harder to read than one with two.
Start and end events	How many of each	Each extra start or end multiplies the cases a reader must consider.
Nesting depth	How many levels of sub-processes sit inside each other	Deep nesting means jumping between many diagrams.
Control-flow complexity (CFC)	How many different "states" the gateways can create. An XOR split counts its number of outgoing branches (n). An AND split counts 1. An OR split counts 2^n minus 1.	It reflects the number of decisions a reader must keep in mind. It was proposed by Cardoso.

Paper 1 reports that size and control-flow complexity are the measures used most often in the empirical studies it reviewed. That makes them sensible first choices for us.

A worked example

Take an online shop that handles an order. Figure 2 draws it cleanly. Figure 3 draws the same business goal with several of the habits from section 1.5. The second diagram is made up for this document to show the effect, and it is not meant to match the first one step for step.

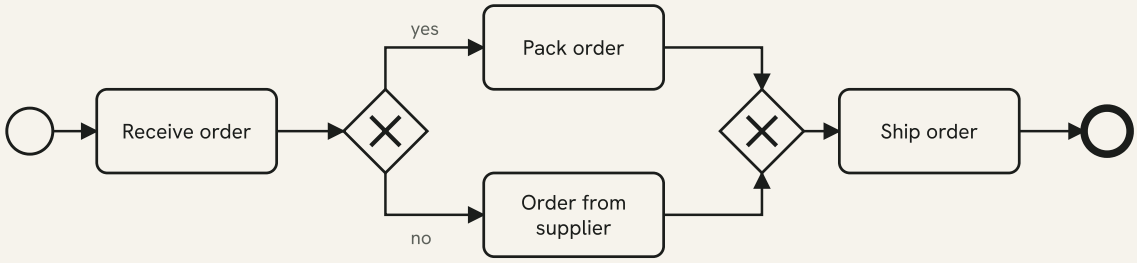


Figure 2. Version A. A clean model: one start, one end, one split and one matching join. The split asks whether the item is in stock (yes or no).

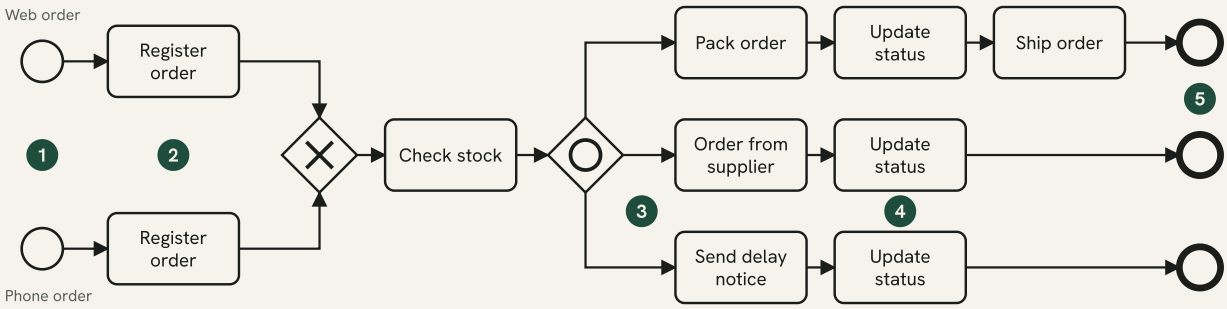


Figure 3. Version B. Problems marked with numbers. 1: two start events. 2: "Register order" drawn twice. 3: an OR split that is never joined again. 4: "Update status" repeated three times. 5: three end events.

Metrics for the two versions

METRIC	VERSION A	VERSION B
Nodes (events, tasks, gateways)	8	17
Arcs (arrows)	8	16
Start events	1	2
End events	1	3
Largest fan-out at one gateway	2	3
Repeated task copies	0	3
Control-flow complexity (CFC)	2	7
Guidelines from Paper 3 that are broken	0 of 7	4 of 7 (G1, G3, G4, G5)

How the CFC numbers come out. Version A has one XOR split with 2 branches, so CFC is 2. Version B has one OR split with 3 branches, so CFC is 2^3 minus 1, which is 7. Just by swapping an XOR for an OR and adding one branch, the measured complexity more than triples.

A single number never settles the question. A high value tells you where to look. Then a person decides whether the complexity is essential or avoidable.

Detection in process mining

Process mining builds a model automatically from event logs, which are records of what actually happened in a system. These models are often very large and tangled. Paper 2 reviews this problem. Its authors point out that the field has no agreed definition of complexity and no standard way to measure it, so existing solutions are mostly ad hoc. They propose a definition, list the factors that contribute to complexity, and catalogue the metrics used.

1.8 Reducing unnecessary complexity

Once a problem is detected, the next step is to fix it. The golden rule is: **make the model simpler without changing what the process does.** If a "fix" quietly changes the behaviour, it is not a reduction, it is a different process.

Main ways to reduce complexity

TECHNIQUE	IDEA	WATCH OUT FOR
Follow modelling guidelines	Draw with rules such as the seven in Paper 3: few elements, one start and one end, matched splits and joins, no OR gateways.	Rules are rules of thumb. Some real processes need to break them.
Restructure the flow	Give every split a matching join of the same type. Merge several start or end events. Replace an OR gateway with XOR or AND where the process allows it.	The new structure must allow exactly the same paths as before.
Merge duplicates	Replace repeated copies of a task or fragment with one shared copy.	Two similar fragments may differ in a small but important way.
Decompose into sub-processes	Hide detail inside collapsed sub-processes so each diagram stays small. Paper 3 suggests this for models with more than 50 elements.	Paper 1 found mixed evidence. One study reported less mental effort with modules but no better task results, and another found understanding got worse as nesting got deeper.
Abstract or simplify	Hide less important detail, or in mined models filter out rare behaviour so the main flow is visible. Paper 2 groups the existing solutions into four approaches.	Some information is lost on purpose, so decide what the reader really needs.
Improve labels and visual cues	Name tasks with a verb and an object. Use colour to highlight matching gateways. Lay out the diagram cleanly.	Paper 1 mentions one study where colouring helped novices on accuracy, but the evidence is thin.

1.9 What is still unsettled

Knowing the limits of a field is a sign of a good project. From the three base papers, these points are still open.

- There is no single accepted definition or measurement of complexity (Paper 2).
- Studies disagree on some factors. Structural complexity has strong agreement, but modularity results are inconclusive, and the authors of Paper 1 call out gaps, inconsistent findings and weak methods in parts of the field.
- Thresholds such as "split models above 50 elements" are guidance drawn from earlier studies, not laws (Paper 3).
- Reducing complexity can hide information, so there is always a trade-off between simple and complete.

1.10 The topic in six lines

- A business process model is a diagram that people use to discuss, improve and automate work.
- A model is only useful if people can read it correctly.
- Some complexity is essential. Some is added by how the model was drawn, and that part is unnecessary.
- Unnecessary complexity makes models harder to understand and more error-prone.
- It can be detected with metrics and rule checks, and reduced by restructuring, merging, decomposing and clarifying.
- Any reduction must keep the process behaviour the same.

PART 2

Problem statement

2.1 How a problem statement is built

A good problem statement answers four questions in order: what is the situation, what goes wrong, why does it matter, and what exactly will we do about it. Below, each answer is first given in plain words, and then joined into one statement you can read aloud.

QUESTION	OUR ANSWER
Situation	Organisations use process models to understand, discuss and automate their work. These models are meant to make a process easier to grasp.
What goes wrong	Many models are larger and more tangled than the process requires. They carry extra start and end events, repeated tasks, unmatched splits, OR gateways and over-sized diagrams.
Why it matters	Readers understand such models less well, errors become more likely, and mined models give little insight. Many modellers are not experts, so they often cannot tell which parts are avoidable.
The gap	The literature offers many metrics and some guidelines, but studies disagree on several points and there is no agreed definition or measurement of complexity. From the papers we read, a modeller still has no simple, combined way to find the avoidable part and fix it safely.

2.2 The problem statement

PROBLEM STATEMENT

Business process models are meant to make a process easy to understand, yet in practice many of them contain more elements and more tangled structure than the underlying process needs. This unnecessary complexity makes models harder to read, more likely to contain errors, and costly to maintain. Existing research offers many complexity metrics and a number of modelling guidelines, but they are scattered, partly inconsistent, and rarely combined into a practical way to tell avoidable complexity from essential complexity and to remove it without changing the process behaviour.

This project studies how unnecessary complexity in business process models can be detected using established metrics and guideline checks, how it can be reduced through behaviour-preserving restructuring, and how much these steps improve the models.

The wording is a draft for you and your teammate. Keep the structure, change words to your voice, and check the scope with your professor.

2.3 Objectives

1. **Collect** from the literature the complexity metrics and modelling guidelines that are most used and easiest to compute.
2. **Detect** unnecessary complexity in a sample of BPMN models by computing metrics and checking guideline violations.
3. **Reduce** it using simple, behaviour-preserving changes, and record exactly what was changed.
4. **Evaluate** by comparing metric values and guideline violations before and after, and optionally by asking classmates to read both versions.

2.4 Research questions

NO.	QUESTION	HOW WE ANSWER IT
RQ1	Which complexity metrics and guideline checks are most common in the literature, and which of them can be computed automatically on BPMN models?	Reading the base papers, then a short script.
RQ2	How much avoidable complexity do typical BPMN models contain when measured with these metrics and checks?	Running the script on a sample of models.
RQ3	Do simple reductions (matching splits and joins, merging duplicates, decomposing) lower the measurements while keeping the process behaviour the same?	Before and after comparison, with a manual behaviour check.
RQ4	Optional: do readers find the reduced models easier to understand?	A short reading task with a few classmates.

2.5 Scope: what is in and what is out

In scope

- BPMN models, looking at control flow only: events, tasks and gateways.
- A small sample of models from textbooks, courses or public collections.
- Metrics and rule checks taken from the base papers.
- Simple reductions that keep the same behaviour.

Out of scope

- Inventing a brand new metric.
- Data objects, resources, time and cost in the models.
- Building a full process mining algorithm.
- Proving formal behaviour equivalence for every model.

2.6 A realistic way to do it

1. **Choose 10 to 20 BPMN models.** Use ones from your course, a textbook, or a public collection of process models. Include a mix of small and large ones.
2. **Export them as BPMN files.** Tools such as the free bpmn.io or Camunda Modeler can open and save the standard format, which is plain XML.
3. **Write a small script, for example in Python, that reads the file** and computes size, arcs, gateway mix, largest fan-out, start and end events, and control-flow complexity. Add checks for the seven guidelines that can be tested automatically.
4. **Rank the models and read the worst ones by hand.** Decide which complexity is essential and which is avoidable.

5 **Apply reductions** to the avoidable parts, one change at a time, and note each change.

6 **Measure again** and fill a before and after table like the one in section 1.7.

7 **Report limits honestly,** for example that a number cannot tell essential from avoidable complexity on its own.

This plan is only a suggestion to make the problem statement concrete. For the current submission you only need the topic, the problem statement and the papers.

2.7 The statement in 30 seconds

Process models should make work easier to understand, but many are more tangled than the process needs. We want to detect that extra complexity with known metrics and rules, remove it without changing how the process works, and measure whether it helps.

PART 3

Base papers and references

3.1 How the papers were chosen

You asked for papers that are not too old, are well cited, and are not too hard for a 3rd year student meeting the subject for the first time. Candidate papers were searched on the topic, and their year, venue and citation counts were checked in the OpenAlex database on 21 October 2026. The abstracts of the three chosen papers were then read.

Selection criteria

CRITERION	WHAT WAS APPLIED
Relevant	Must cover complexity of process models, how to detect it, or how to reduce it.
Recent enough	Prefer 2017 onwards. One older classic is kept because the field's practical rules come from it.
Cited	Published in a recognised journal and cited by other researchers. No paper with near-zero citations.
Readable	Review papers and guideline papers, not heavy formal mathematics.

Citation counts here come from OpenAlex. Google Scholar usually shows higher numbers for the same paper, so do not be surprised if you see a different figure when you check.

3.2 The three base papers at a glance

NO.	PAPER	YEAR	CITED BY	ROLE IN OUR PROJECT
P1	Factors influencing the understandability of process models: A systematic literature review	2018	96	Why complexity hurts, and what evidence exists
P2	Complex process modeling in process mining: A systematic review	2022	23	How complexity is defined, measured and tackled
P3	Seven process modeling guidelines (7PMG)	2010	626	Practical rules to reduce complexity

On the year: Papers 1 and 2 are recent. Paper 3 is from 2010, which is older than you asked for. It is kept as a base paper because it is the most cited practical guideline paper in this area, and it is short and easy. If your professor insists on

newer papers, replace it with the 2017 metrics review listed in section 3.7 and keep Paper 3 as background reading.

Suggested reading order: Paper 3 first (10 pages, easiest), then Paper 1 (18 pages), then Paper 2 (22 pages).

3.3 PAPER 1

Factors influencing the understandability of process models: A systematic literature review

Ahmet Dikici, Oktay Türetken and Onur Demirörs. Information and Software Technology, volume 93, pages 112 to 129, 2018. DOI 10.1016/j.infsof.2017.09.001

TYPE	Systematic literature review. The authors collect many earlier studies with a fixed method and summarise them.
SIZE OF THE REVIEW	Studies from 1995 to 2015 in six electronic libraries. 1,066 publications were found at first and 45 primary studies were kept.
READING EFFORT	Medium. 18 pages, with tables of studies. Read the abstract, introduction, results and conclusion.

What it says, in simple words

The authors asked: what makes a process model easy or hard to understand? They found 20 factors and sorted them into two groups. Twelve are factors of the model itself, such as notation, structural complexity, modularity, layout, labels and colour. Eight are factors of the reader, such as modelling experience, knowledge of the notation and familiarity with the domain.

For our topic the key result is about structural complexity. It has the strongest agreement among the studies: more complex structure makes models harder to understand. The two measures used most often are control-flow complexity and size. On modularity the evidence is mixed. The authors also note gaps, inconsistent findings and weak methods in parts of the field.

How we use it. It is our main evidence that complexity is a real problem, and it tells us which measures researchers actually use. Its list of open gaps supports the "gap" in our problem statement.

Free author copy:
pure.tue.nl/ws/portalfiles/portal/78851660/2017_IST_Dikici_Turetken_AuthorVersion.pdf

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3.4 PAPER 2

Complex process modeling in process mining: A systematic review

Mohammad Imran, Maizatul Akmar Ismail, Suraya Hamid and Mohammad Hairul Nizam Md Nasir. IEEE Access, volume 10, pages 101515 to 101536, 2022. DOI 10.1109/ACCESS.2022.3208231

TYPE	Systematic literature review, published in an open access journal.
SIZE OF THE REVIEW	Six major scholarly databases were searched.

READING EFFORT Medium to hard. 22 pages. It assumes a little process mining knowledge. Skip the algorithm details and focus on the definition of complexity, the metrics and the taxonomy.

What it says, in simple words

Process mining discovers a model from event logs. Real processes are often unstructured, so the discovered models tend to be complex and give little useful insight. Many methods try to fix this, but the field has no explicit definition of complexity and no standard way to measure it, so the methods are ad hoc and hard to compare.

The authors review the literature to clear this up. They propose an explicit definition of complexity, identify what contributes to it, catalogue the complexity metrics and where each applies, and organise the solutions into a taxonomy of four approaches. They close with open challenges and future directions.

How we use it. It is our most recent paper and it speaks directly to "detecting" (definition and metrics) and "reducing" (the four approaches). One caution: it is about models discovered from logs, while our project works on hand-drawn BPMN models. We borrow its definitions, metrics and ideas, and we do not need its mining algorithms.

3.5 PAPER 3

Seven process modeling guidelines (7PMG)

Jan Mendling, Hajo A. Reijers and Wil M. P. van der Aalst. Information and Software Technology, volume 52, issue 2, pages 127 to 136, 2010. DOI 10.1016/j.infsof.2009.08.004

TYPE Guideline paper built on earlier empirical studies of how people understand models and where errors occur.

READING EFFORT Easy. 10 pages. Read all of it, especially the list of seven guidelines.

What it says, in simple words

Model quality has had little rigorous study in practice, and existing quality frameworks are too abstract for casual modellers. The authors gather what research says about model structure, understanding and errors, and turn it into seven short guidelines that are both evidence-based and practical.

The seven guidelines, paraphrased

NO.	GUIDELINE
G1	Use as few elements as possible. Bigger models are harder to read and more error-prone.
G2	Keep the number of paths going in and out of a single element small.
G3	Use one start event and one end event.
G4	Keep the model structured: every split gets a matching join of the same type.
G5	Avoid OR gateways. Models using only AND and XOR have fewer errors.
G6	Label tasks with a verb and an object, such as "Inform customer".
G7	Split a model with more than 50 elements into smaller, well-defined sub-models.

How we use it. It is our checklist for detection and our menu for reduction. Several guidelines are easy for a script to test (G1, G2, G3, G5 and G7). G4 needs a little more work and G6 needs word analysis. Together they give us a clear, mostly automatic rule check. It also supplied the example in section 1.7.

Free copy: www.vdaalst.com/publications/p574.pdf

3.6 How the three papers fit together

QUESTION IN OUR PROJECT	PAPER THAT HELPS
Is unnecessary complexity a real problem?	P1 shows that structural complexity harms understanding. P3 links size and tangle to errors. P2 shows it makes mined models unhelpful.
How do we detect it?	P2 gives a definition and a catalogue of metrics. P1 shows which measures are used most. P3 gives rule checks.
How do we reduce it?	P3 gives guidelines to follow. P1 gives cautions about modularity and colour. P2 gives the four approaches.
What is the gap?	P2 notes no standard definition or measure. P1 notes inconsistent findings. P3 notes that practical, evidence-based advice for casual modellers was missing.

3.7 Extra reading

These papers are good backups or add-ons. For them, only the title, venue and citation count were checked, and their abstracts were not read for this document, so skim each one before you rely on it.

PAPER	YEAR	CITED BY	WHY IT MAY HELP
Polančič and Cegnar. Complexity metrics for process models, a systematic literature review. Computer Standards and Interfaces, 51, 104-117.	2017	49	Review of complexity metrics. Best swap if you need a newer paper than P3.
Smirnov, Reijers, Weske and Nugteren. Business process model abstraction: a definition, catalog, and survey. Distributed and Parallel Databases.	2012	122	Techniques for reducing a model by abstraction.
Figl. Comprehension of procedural visual business process models: a literature review. Business and Information Systems Engineering.	2017	140	How people read process models, useful for RQ4.
Weber, Reichert, Mendling and Reijers. Refactoring large process model repositories. Computers in Industry.	2011	220	Finding and removing repeated parts across many models.

3.8 References

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The control-flow complexity metric in section 1.7 is credited to Cardoso, whose original paper was not read for this document.

PART 4

Presenting this

4.1 Slide plan for 8 to 10 minutes

NO.	SLIDE TITLE	WHAT GOES ON IT	SPEAKER
1	Topic and team	Title, course, names.	Person 1
2	A process model is a diagram of work	Figure 1 and one small example such as the pizza order.	Person 1
3	Some complexity is not needed	Figures 2 and 3 side by side, with the five marked problems.	Person 1
4	Complex models cost readers and accuracy	Three bullets: harder to understand (P1), more errors (P3), unhelpful mined models (P2).	Person 2
5	Our problem and our questions	The problem statement in two sentences and the four research questions.	Person 2
6	Three base papers	The table from section 3.2 with one line on what each gives us.	Person 2
7	Plan and next steps	The seven-step approach from section 2.6, shortened to four bullets.	Person 2

Keep the slides sparse. Put the figures on the slides and say the explanations aloud. Use the metrics table from section 1.7 as your "proof" slide if your professor likes numbers.

4.2 Questions a professor may ask

QUESTION	A GOOD SHORT ANSWER
How is unnecessary complexity different from a complex process?	A complex process needs a complex model. Unnecessary complexity is extra tangle added by how we draw the model, such as duplicate tasks or unmatched splits, which can be removed without changing the behaviour.
How will you measure it?	With metrics from the literature, mainly size, gateway counts, fan-out and control-flow complexity, plus automatic checks of the seven guidelines.
Won't simplifying lose information?	It can, which is why we only apply changes that keep the same paths, we note every change, and we report this trade-off as a limit.
Why these papers?	One shows the evidence that complexity hurts understanding, one defines and measures complexity and reviews solutions, and one gives practical rules. Two are recent reviews and one is the most cited guideline paper.
What is new in your project?	We do not claim a new theory. We bring existing metrics and rules together in one small, runnable check and test whether the suggested fixes really reduce complexity on a sample of models.
Why is Paper 2 about process mining?	Because it has the clearest discussion of how complexity is defined and measured. We use its ideas, and our models are drawn by hand in BPMN.

4.3 Glossary

TERM	PLAIN MEANING
Business process	A series of steps carried out to reach a business goal.
BPMN	The standard drawing language for process models.
Event	Something that starts, ends or interrupts a process.
Gateway	A diamond where paths split or join.
Split and join	A split sends flow into several paths. A join brings paths back together.
XOR, AND, OR	Exactly one path, all paths, or one or more paths.
Sub-process	A box that hides a smaller process inside it.
Control flow	The order in which tasks can happen.
Metric	A number computed from a model, such as its size.
Spaghetti model	A model with tangled, hard-to-follow flow.
Process mining	Building process models automatically from system logs.
Event log	A record of what happened in a system, step by step.
Systematic literature review	A paper that collects and summarises many studies using a fixed, repeatable method.
Understandability	How easily and correctly people can read a model.